Decimal	Binary $w\ x\ y\ z$	Term
8, 9, 10, 11 (1, 2)	1 0 – –	wx'

• The sum of the prime-implicants gives a valid algebraic expression for the function. However, this expression is not necessarily the one with the minimum number of terms.

$wx \backslash yz$	00	01	11	10
00		1		
01	1		1	1
11			1	
10	1	1	1	1

$F = wx' + xyz + w'xz' + x'y'z$

3.11 Selection of Prime-implicants

The selection of prime implicants that form the minimized function is made from a prime-implicant table. In this table, each prime-implicant is represented in a row and each minterm in a column. Crosses are placed in each row to show the composition of minterms that make the prime-implicants.

Table 3-8 Prime-implicant table for Example 3-15

		1	4	6	7	8	9	10	11	15
√ $x'y'z$	1, 9	X					X			
√ $w'xz'$	4, 6		X	X						
$w'xy$	6, 7			X	X					
xyz	7, 15				X					X
wyz	11, 15								X	X
√ wx'	8, 9, 10, 11					X	X	X	X	
		√	√	√		√	√	√	√	

• There are six rows, one for each prime-implicant (derived in Example 3-14), and nine columns, each representing one minterm of the function. Crosses are placed in each row to indicate the minterms contained in the prime-implicant of that row. For example, the two crosses in the first row indica